

Finding Maclaurin's series

Key Points

To find the Maclaurin's series for any (differentiable) function $f(x)$ differentiate and evaluate at $x = 0$ as required for use in the formula

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

For equality we need infinite terms, but we can use any amount of terms as an approximate

Basic Skills

Q1. Differentiate the following functions

a) $y = 4x^2 - 5$

b) $y = \cos 2x$

c) $y = e^x \sin x$

d) $y = \tan x^2$

e) $y = \frac{x^2}{\ln x}$

f) $y = \ln^2 x$

Q2. Find the gradient of the function $f(x) = \sin x$ at the origin

Q3. Find the stationary points and classify them for the function $f(x) = x^2 e^{5x}$

Q4. Find the first two derivatives $f'(x), f''(x)$ of the function $f(x) = \sqrt{8 + e^x}$

Competency Skills

Q5. For the function $f(x) = \cos(x^2)$

a) Find the derivatives $f'(x), f''(x)$

b) Hence find $f'(0), f''(0)$

c) Hence find the Maclaurin's series for $f(x)$ up to the x^2 term

Q6. For the following functions find the Maclaurin's series up to the x^2 term

a) $f(x) = e^{3x}$

b) $f(x) = \tan x$

c) $f(x) = \sqrt{1 + x^2}$

Q7. Hence using your answers to Q6., other otherwise, find the first 3 terms of the Maclaurin's series for $f(x) = xe^{3x}$

Advanced Skills

Q8. Find the Maclaurin's series for $f(x) = \sin e^x$ up to the x^2 term

Q9. Find a linear approximation of $f(x) = \arctan(1 + x)$

Q10. Explain the idea behind a Maclaurin's series, including why we might calculate one

Q11. By using the first three non-zero terms of $\cos x$'s Maclaurin's series find an estimate for $\cos \pi$, commenting on the accuracy of the estimate

Q12. By finding the Maclaurin's series for $f(x) = \frac{1}{1-x}$ and $g(x) = \ln(1 + x)$ find the Maclaurin's series for $h(x) = \frac{\ln(1+x)}{1-x}$ up to the x^3 term

Extension

Q13. What difference would evaluating the function and it's derivatives at $x = a$ for $a \neq 0$ make? What would the new series now do compared to Maclaurin's

Hint: look up Taylor's series

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